

restart

$$edo := \frac{d^2}{dx^2} y(x) + 0.199809 y(x) = 0$$

$$cond := y(0) = 1, D(y)(0) = 2$$

$$sol := y(x) = \frac{2000}{447} \sin\left(\frac{447}{1000} x\right) + \cos\left(\frac{447}{1000} x\right)$$

$$des := \frac{2000}{447} \sin\left(\frac{447}{1000} x\right) + \cos\left(\frac{447}{1000} x\right)$$

$$vel := 2 \cos\left(\frac{447}{1000} x\right) - \frac{447}{1000} \sin\left(\frac{447}{1000} x\right)$$

$$acel := -\frac{447}{500} \sin\left(\frac{447}{1000} x\right) - \frac{199809}{1000000} \cos\left(\frac{447}{1000} x\right)$$

$$ur := 2.4504$$

$$vmax := 2.05$$

$$amax := 0.916$$

